Model Interpretability: Understanding Neural Network Decisions

Wednesday, 21 October 2026 7:33 PM

Agenda:

- Business Case
- LIME
  - Global vs. Local Analysis
  - Locality of Interest
  - Model Agnosticism
  - Code Implementation

A model to predict if a patient has cancer or not → Yes/No.

A model to predict if a customer should be given loan or Not → Yes/No

C_1 → No

Why No? → How to answer?

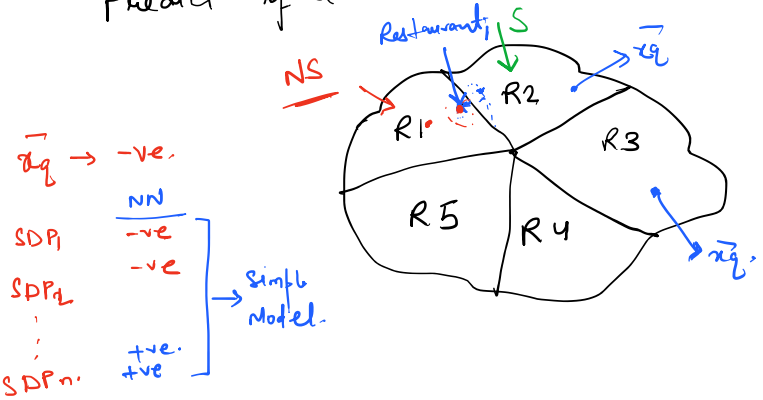
Accuracy is 95%.

— NN is considered a black box.

↓

can have millions of W & B.

Predict if a restaurant is servicable or not.

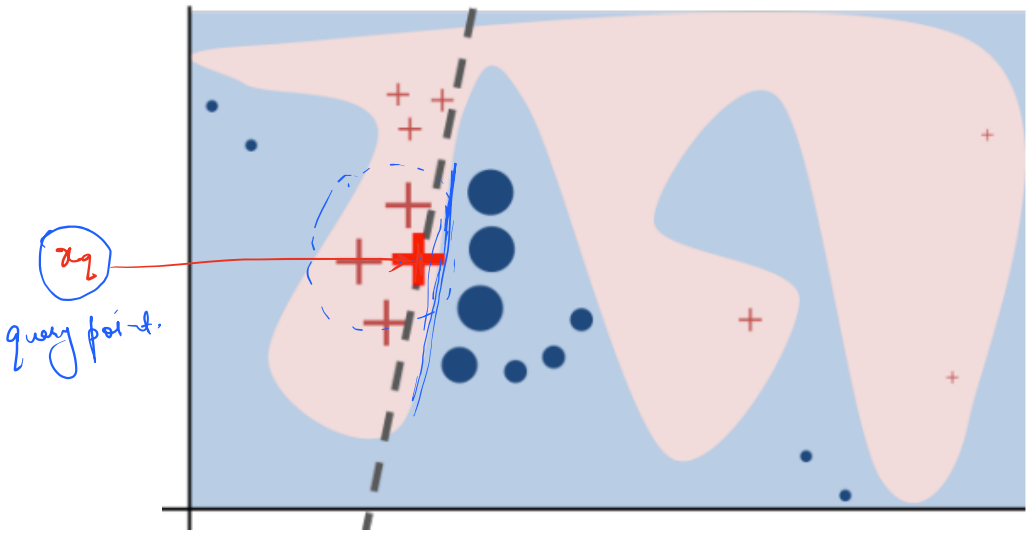


at 15 mins interval we used to predict servicability.

* We need model explainability to explain the predictions of a model.

— LIME → Today's Class.

— SHAP → Next Class.



L
↓
Local

* Explain a single data point.

I
↓
Interpretable

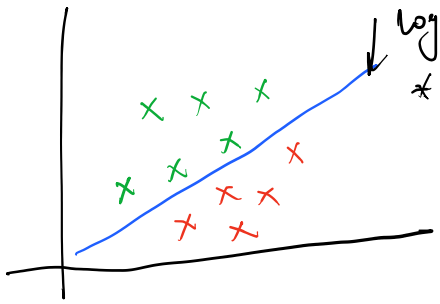
* Explain

M
↓
Model Agnostic

- NN
- Decision Trees → XGBoost, RF
- Any model

E
↓
Explanation

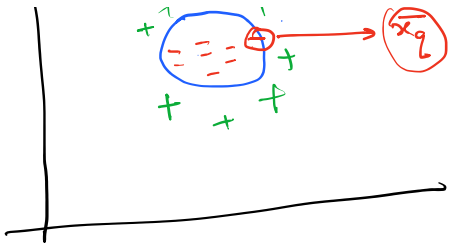
* Global Analysis



* Decision boundary is based on all data points.

* Local Analysis

... to the decision



Local Analysis → what is special about $\bar{x}_q$ point and how varying this data point caused the prediction to change.

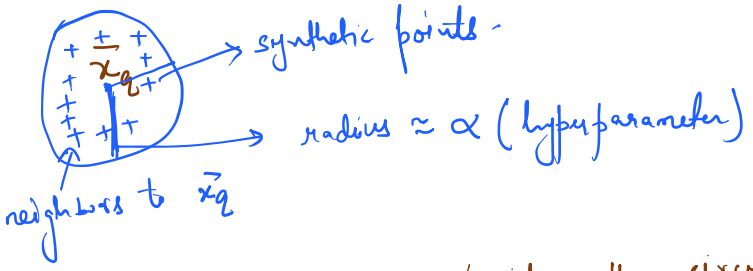
* Locality of Interest

how can I change data point $\bar{x}_q$? → 1 customer.

$$\bar{x} = \bar{x}_q + \varepsilon_i \Big\}_{i=1}^{10,000}$$

↑ adding noise

create $N = 10,000$ datapoints which will be neighbors to $\bar{x}_q$.



all new/synthetic data points will be inside the circular region with radius α.

* Model Agnostic.

Linear models → weights

Decision trees → conditions

... models → different techniques.

for different models

LIME → General technique.

It can be applied to any model.

model agnostic.

* Steps of LIME

1) Neural Network (Any model) is trained, evaluated, prediction. ✓

2) Select a local point $\vec{x_q}$ → A person for which we want to understand the prediction.

3) Create N data points within the α radii of $\vec{x_q}$ by introducing ε noise to the local point $\vec{x_q}$

4) Create a dataset with new data points.

Step 3 →

x	y
SDP_1	$NN(SDP1)$
SDP_2	$NN(SDP2)$
SDP_3	$NN(SDP_3)$
⋮	⋮
SDP_n	$NN(SDP_n)$

(NN ← main model)

5) Build a surrogate model (Lasso Model)

↓

simple / interpretable model → Un Reg / Log Reg.

fit it on the synthetic data points

→ $\vec{x_q}$

+ve, -ve.

feature imp.

... to explain the prediction

Take the top features from this model to explain on $\bar{x_q}$.

1 DP → Local explanation.

all DP → Global explanation.

A hospital wants to know which single input most often drives the model's diabetes predictions across all ten thousand patients in its database. Which kind of explanation answers this?

Global Explanation.